

# IndusBoard Standalone IoT LIDAR RADAR

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**L**idar plays a key role in cartography, mapping, localisation, ADAS, environment scanning, and more. It is widely used in robots, autonomous vehicles, and for intruder monitoring. However, many

systems are wired, making them cumbersome for prototyping and testing, particularly in autonomous driving, where mounting and connecting a lidar can be difficult.

Here, we present a Wi-Fi-enabled

standalone lidar that scans and delivers real-time data wirelessly. Simply connect it to a battery, and it is ready to use—no wiring required.

This device enables flexible lidar deployment for mapping, monitor-

**TABLE 1**  
**BILL OF MATERIALS**

| Components             | Description                 | Quantity |
|------------------------|-----------------------------|----------|
| IndusBoard             | 3cm sized development board | 1        |
| 360 degree lidar       | YD X2 lidar                 | 1        |
| 5V battery             | Power source                | 1        |
| 3.3V voltage regulator | DC-DC 5V to 3.3V regulator  | 1        |

**TABLE 2**  
**PIN MAPPING**

| Pin   | Type         | Description                  | Defaults | Range     | Remarks                                          |
|-------|--------------|------------------------------|----------|-----------|--------------------------------------------------|
| VCC   | Power supply | Positive                     | 5V       | 4.8V-5.2V | /                                                |
| Tx    | Output       | System serial port output    | /        | /         | Data stream: Lidar to peripherals                |
| GND   | Power supply | Negative                     | 0V       | 0V        | /                                                |
| M_CTR | Input        | Motor speed control terminal | 1.8V     | 0V-3.3V   | Voltage speed regulation or PWM speed regulation |



Fig. 1: Fixing IndusBoard onto the lidar



Fig. 2: YD X2 lidar

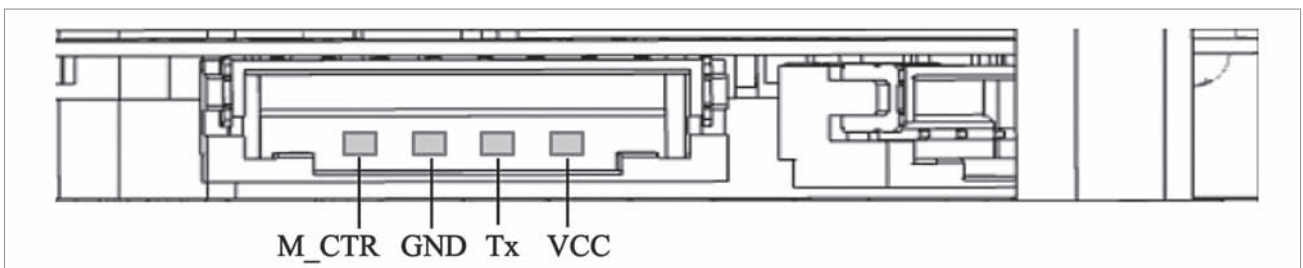


Fig. 3: Lidar connector pins (Source: YD X2 lidar datasheet)